

Chapter 3 — Introduction to ethics and privacy

Chapter 2 asked how records enter a dataset

Learning objectives

- Distinguish ethics from privacy, name stakeholders affected across the dataset lifecycle
- Explain why compliance alone does not guarantee ethical practice

Why ethics and privacy matter now

- Data-driven systems shape decisions in healthcare, finance, marketing, and government
- As collection and use scale
- Misuse can enable identity theft, fraud, discrimination

Defining ethics and privacy

- In data work, ethics refers to principles that guide how data are collected, used, shared
- Privacy denotes individuals' rights to control personal data
- Ethics asks whether a use should proceed

Ethics versus privacy at a glance

- Ethics focuses on who is harmed or helped and whether uses are fair and accountable
- Privacy focuses on meaningful consent and individual control
- A team can be legally compliant yet ethically weak if uses are opaque or discriminatory

Lifecycle and stakeholders

- Documentation
- Stakeholders include practitioners, leaders, policymakers
- No single role owns the full obligation

Example 3.1 — Stakeholders in a wellness app launch

- Example 3.1 — hands-on module
- Example 3.1 compresses those roles into one product decision
- Product wants finer activity tracking
- Ethical design must negotiate utility, consent, and minimization together
- Explore the chapter example module
- View files: ``modules/chapter3/example1/``

Takeaways

- Ethics and privacy are distinct but mutually reinforcing
- Obligations span the full dataset lifecycle, not only deployment
- Ethical data work is coordinated across roles, utility, consent

Next

- Complete the quiz for this part
- Continue to the next part on core ethical principles

Learning objectives

- Name the five core principles, state what each demands in practice
- Recognize them as design constraints, not post-deployment patches

Five principles overview

- Non-maleficence asks teams to prevent privacy breaches, discrimination
- Fairness requires avoiding systematic disadvantage and monitoring disparities
- Transparency means disclosing collection purposes and
- Accountability assigns responsibility, monitoring, and remediation
- Respect for autonomy centers informed consent and ongoing control

Do no harm

- Do no harm requires anticipating harms from collection, processing
- Yielding less accurate predictions and misdiagnosis risk for that group
- Upholding non-maleficence means auditing subgroup performance and adding safeguards before

Example 3.2 — Healthcare AI system

- Example 3.2 — hands-on module
- The clinical pattern: skewed training data becomes skewed care
- Explore the chapter example module
- View files: ``modules/chapter3/example2/``

Fairness

- Fairness requires that practices and models not systematically disadvantage particular
- The model can reproduce that pattern for new applicants
- Treat disparity risk as a design constraint, not a post-deployment discovery

Example 3.3 — Hiring algorithm

- Example 3.3 — hands-on module
- Example 3.3 restates the hiring pattern
- Chapter 7 develops metrics and mitigation
- Explore the chapter example module
- View files: ``modules/chapter3/example3/``

Transparency and accountability

- Transparency builds trust by clarifying what is collected, why
- Accountability assigns someone who can explain outcomes, respond to incidents
- Example 3.4 applies transparency to credit scoring

Respect for autonomy

- Respect for autonomy centers informed consent and ongoing control
- Lets users revoke consent or delete history
- Autonomy fails when consent is buried in unrelated terms or withdrawal is practically

Example 3.6 — Health-tracking app

- Example 3.6 — hands-on module
- Example 3.6 pairs consent user experience with sensitive streams such as activity and
- Explore the chapter example module
- View files: ``modules/chapter3/example6/``

Takeaways

- Principles are operational tests for design choices
- Non-maleficence and fairness catch harm before deployment
- Transparency and accountability make systems answerable
- Autonomy requires meaningful consent and control, not checkbox compliance alone

Next

- Complete the quiz for this part
- Continue to utility trade-offs, ethical dilemmas when principles conflict

Learning objectives

- Explain the utility–privacy tension, describe two ethical dilemma patterns
- List consequences of ignoring ethics for people and organizations

Balancing utility and privacy

- Personalized care
- Treat statutes as floors rather than ceilings
- The ethical task is deciding whether proposed utility warrants the privacy burden at all

Example 3.7 — Utility–privacy tension in personal predictions

- Example 3.7 — hands-on module
- Example 3.7 states the trade-off in prediction settings
- Models that score health risk or creditworthiness can improve service allocation
- Without minimization, access control
- Explore the chapter example module
- View files: ``modules/chapter3/example7/``

Ethical dilemmas in practice

- Dilemmas arise when principles conflict or no option is clearly harm-free
- Example 3.8 sketches criminal-justice risk scores
- Example 3.9 returns to platform collection

Example 3.8 — Dilemma in criminal justice risk scores

- Example 3.8 — hands-on module
- The justice-system conflict previewed again in the bias section and Chapter 7
- More data alone does not dissolve the conflict
- Explore the chapter example module
- View files: ``modules/chapter3/example8/``

Consequences of neglecting ethics

- Ignoring ethics produces privacy violations, discrimination, eroded trust
- Example 3.10 shows personal exposure after unauthorized use or leakage, fraud, harassment

Example 3.10 — Misuse of personal data

- Example 3.10 — hands-on module
- Example 3.10 connects ethical neglect to breach economics developed later in the chapter
- Organizations also face fines, litigation
- Ethical practice is both a moral and a sustainability requirement
- View files: ``modules/chapter3/example10/``

Takeaways

- Utility and privacy trade off by design, not by accident
- Dilemmas need structured decision procedures, not silent defaults
- Neglecting ethics harms people first and balance sheets second

Next

- Complete the quiz for this part
- Continue to data privacy principles, consent, anonymization

Learning objectives

- Define data privacy in pipeline terms, contrast three core privacy-management concepts
- Read GDPR versus HIPAA as planning maps, not compliance manuals

From rights to management practice

- Section 3.1 defined privacy as individuals' control over personal data
- This section asks which agreements and defaults make that control enforceable
- Personal data includes names and addresses but also behavioral logs
- Ethics still decides whether a use should proceed

Three core concepts

- Informed consent requires freely given
- Anonymization reduces linkability for secondary use
- Privacy by Design embeds protections into architecture from the start rather than bolting

Informed consent in healthcare

- Informed consent is the operational counterpart of respect for autonomy
- Example 3.11 shows a clinic that wants to reuse treatment records for research
- Bundling research into a general intake form fails the standard

Example 3.11 — Consent for healthcare data uses

- Example 3.11 — hands-on module
- The clinical consent bar
- Explore the chapter example module
- View files: ``modules/chapter3/example11/``

Privacy by Design

- Privacy by Design means minimization, protective defaults
- Should expose clear sharing and deletion controls before any partner sync

Example 3.12 — Privacy by Design in a health app

- Example 3.12 — hands-on module
- Example 3.12 pairs with the autonomy vignette in Example 3.6 but focuses on architecture
- Explore the chapter example module
- View files: ``modules/chapter3/example12/``

GDPR and HIPAA as anchors

- The GDPR governs personal data of people in the EU and EEA with extraterritorial reach for
- It emphasizes lawfulness, purpose limitation, minimization, accountability
- HIPAA concentrates parallel duties on protected health information held by US covered
- Section 3.6 expands scope, rights, and penalties

Takeaways

- Privacy management is more than notices
- Consent must be meaningful
- GDPR and HIPAA illustrate how principles become enforceable duties

Next

- Complete the quiz for this part
- Continue to privacy-preserving techniques, anonymization, encryption, differential privacy

Learning objectives

- Place four techniques on a protection map, state the main caveat for each
- Explain how they stack with consent and purpose limits

Techniques at a glance

- Anonymization and de-identification reduce linkability in shared tables
- Encryption protects confidentiality in transit and at rest
- Differential privacy limits how much any single record can change an aggregate release
- Federated learning trains models where raw data stay on devices or silos

Anonymization and re-identification risk

- Coarsening rare combinations
- The canonical warning
- Stronger protection usually requires coarser aggregation

Example 3.13 — Re-identification from location patterns

- Example 3.13 — hands-on module
- Example 3.13 shows why anonymization is a management goal, not a guarantee
- Explore the chapter example module
- View files: ``modules/chapter3/example13/``

Encryption limits

- Encryption keeps intercepted packets unreadable
- Minimization, not from encryption alone
- Example 3.14 makes that limit explicit for medical messaging

Differential privacy

- Differential privacy limits what an adversary learns about any individual from aggregate
- The standing trade-off is privacy budget versus accuracy
- Example 3.15 shows product telemetry released under a differential-privacy mechanism so

Example 3.15 — Differential privacy for product telemetry

- Example 3.15 — hands-on module
- Example 3.15 depends on correct noise calibration and on not publishing too many
- Explore the chapter example module
- View files: ``modules/chapter3/example15/``

Federated learning

- Federated learning reduces central pools of raw text or sensor data by training locally
- Example 3.16 shows on-device keyboard prediction

Example 3.16 — Federated learning for on-device keyboard prediction

- Example 3.16 — hands-on module
- Example 3.16 illustrates when raw centralization is unacceptable
- Explore the chapter example module
- View files: ``modules/chapter3/example16/``

Takeaways

- Choose controls to match the threat
- Stack techniques where needed
- Technical PETs implement privacy principles, they do not replace them

Next

- Complete the quiz for this part
- Continue to bias and fairness, how ethical principles connect to model outcomes

Learning objectives

- Define bias in data work
- List five mitigation habits at the ethics-chapter level

What bias means here

- Bias means systematic deviations in collection
- Data are not neutral mirrors
- Example 3.17 states the general risk

Example 3.17 — Bias in AI system

- Example 3.17 — hands-on module
- Example 3.17 highlights domains where errors change life chances, hiring, criminal justice
- Explore the chapter example module
- View files: ``modules/chapter3/example17/``

Selection bias

- So error concentrates on underrepresented groups
- Collection design in Chapter 2 is often where skew begins
- Example 3.18 shows a diagnostic model trained mainly on one demographic slice

Example 3.18 — Selection bias in healthcare AI

- Example 3.18 — hands-on module
- Example 3.18 relates to the non-maleficence vignette in Example 3.2
- Explore the chapter example module
- View files: ``modules/chapter3/example18/``

Algorithmic bias

- Thresholds can still allocate error unevenly
- Example 3.19 restates hiring rankers that reproduce gender or racial patterns via proxies

Case study pointer — COMPAS

- Example 3.20 points to COMPAS recidivism tools
- The ethics lesson matches Example 3.8, overall accuracy is not a fairness certificate
- Chapter 7 develops detection and impact analysis for this case

Mitigation checklist

- Addressing bias requires better data, explicit fairness criteria
- At this chapter's level

Takeaways

- Fairness work is continuous governance after the principles in Section 3.2
- Selection and algorithmic bias are frequent entry points
- Landmark cases illustrate mechanisms

Next

- Complete the quiz for this part
- Continue to real-world privacy laws, GDPR, HIPAA, CCPA, and selected other regimes

Learning objectives

- Summarize scope and principle highlights for GDPR, HIPAA
- CCPA/CPRA, name data-subject or patient rights teams must honor
- Recognize why global products need jurisdiction maps

From principles to enforceable duties

- Section 3.3 previewed GDPR and HIPAA at principle level
- Details change with amendments, treat the text as a planning map for data teams

GDPR essentials

- Or monitor, those people
- Accountability with documented controls
- Data-subject rights include access, rectification, erasure, restriction, portability

HIPAA essentials

- HIPAA is a US federal framework for protected health information held by covered entities
- The Privacy Rule limits uses and disclosures
- Rehearsing notification workflows

CCPA and CPRA

- To non-discrimination for exercising rights
- Relative to GDPR

Other notable laws

- COPPA requires verifiable parental consent before collecting personal information from
- Brazil's LGPD and China's PIPL echo GDPR-style themes while remaining distinct enforcement
- Global products need maps of which residents are in scope, where data may be stored

Example 3.28 — GDPR fine ceiling

- Example 3.28 — hands-on module
- Example 3.28 illustrates regulatory exposure
- Explore the chapter example module
- View files: ``modules/chapter3/example28/``

Takeaways

- Statutes encode rights catalogs and accountability regimes, not merely privacy policies
- GDPR, HIPAA, and CCPA/CPRA overlap in themes but differ in scope and posture
- Multistate and multinational operators often adopt the strictest baseline they face while

Next

- Complete the quiz for this part
- Continue to data breaches, definitions, causes, and three landmark failure modes

Learning objectives

- Define a data breach, name common causes
- Distinguish three organizational failure modes illustrated by Equifax, Yahoo
- Target

Understanding data breaches

- Excessive third-party access
- Organizations storing identity
- Statute-level notification clocks assume teams can detect and classify incidents quickly

Example 3.22 — Data breach severity range

- Example 3.22 — hands-on module
- Example 3.22 states the severity range
- Impact scales with sensitivity, volume, dwell time before detection
- Explore the chapter example module
- View files: ``modules/chapter3/example22/``

Equifax — unpatched internet-facing systems

- Example 3.23 describes the 2017 Equifax incident
- The failure mode is delayed patching and weak monitoring on high-value identity stores
- Downstream costs appear in the next part

Example 3.23 — Equifax case study

- Example 3.23 — hands-on module
- The patching-discipline lesson
- Explore the chapter example module
- View files: ``modules/chapter3/example23/``

Yahoo — delayed disclosure

- Example 3.24 covers Yahoo breaches affecting hundreds of millions of accounts
- The failure mode is late accountability to users and markets after compromise at consumer

Target — third-party vendor path

- Example 3.25 describes attackers using third-party vendor access to reach point-of-sale
- The failure mode is vendor risk and malware on retail networks

Takeaways

- Breach risk is organizational as much as cryptographic
- Three patterns recur, unpatched systems, delayed disclosure, and vendor-mediated entry
- Prevention controls mapped to each pattern come next

Next

- Complete the quiz for this part
- Continue to breach consequences and prevention

Learning objectives

- List major consequence categories for organizations and individuals
- Match prevention controls to unpatched systems, delayed disclosure
- Vendor entry

Consequences for organizations

- Financial losses include investigation, notification, credit monitoring, litigation
- Regulatory penalties can reach GDPR ceilings illustrated in Example 3.28
- Reputational damage and operational downtime while systems are rebuilt extend harm beyond

Example 3.26 — Equifax breach costs

- Example 3.26 — hands-on module
- Example 3.26 attaches hundreds of millions of dollars in settlements, remediation
- Delayed patching on high-value identity stores becomes a balance-sheet event
- Explore the chapter example module
- View files: ``modules/chapter3/example26/``

Example 3.27 — Yahoo breach and acquisition impact

- Example 3.27 — hands-on module
- Example 3.27 discusses governance and cybersecurity oversight failures that coincided with
- Explore the chapter example module
- View files: ``modules/chapter3/example27/``

Consequences for individuals

- Individuals whose identifiers leak may face identity theft and fraud for years
- Organizations may suffer operational paralysis while forensic work continues
- Prevention therefore protects people first and enterprise value second

Prevention mapped to failure modes

- Timely patching and monitoring address unpatched internet-facing systems
- Rehearsed detection, escalation, and disclosure address delayed accountability
- Vendor risk reviews and least-privilege third-party access address supply-chain entry
- Data minimization

No single control is sufficient

- Equifax, Yahoo, and Target each failed at a different layer of the stack
- Prevention should map to failure modes from Section 3.7, not only to a generic checklist
- Chapter 1 already named encryption and access control as management practices

Takeaways

- Breaches are balance-sheet and governance events, not only IT incidents
- Map controls to known failure patterns
- Minimization ensures stolen stores contain less that can be abused

Next

- Complete the quiz for this part
- Continue to ethical decision-making, structured lenses for conflicts among principles

Learning objectives

- Describe utilitarian, rights-based, virtue
- Justice reasoning
- Explain why stakeholder review belongs before locking thresholds or sharing defaults

Four lenses for data work

- It formalizes the utility–privacy tension from Section 3.2
- Rights-based reasoning holds that autonomy, privacy
- Virtue ethics asks what an honest, fair
- Justice reasoning inspects whether burdens fall disproportionately on already

Example 3.29 — Healthcare prediction

- Example 3.29 — hands-on module
- Even when individuals did not consent to research reuse
- Which is why rights-based checks are needed in parallel
- Explore the chapter example module
- View files: ``modules/chapter3/example29/``

Rights, virtue, and justice vignettes

- Duties rather than optional costs
- Virtue ethics may require disclosing systematic label gaps rather than selling cleaner
- Justice reasoning may delay hiring ranker launch until subgroup audits show disparities

Example 3.33 — User data collection for marketing

- Example 3.33 — hands-on module
- Example 3.33 applies all four lenses to marketing collection under a vague notice
- Utilitarian views emphasize engagement gains
- Explore the chapter example module
- View files: ``modules/chapter3/example33/``

Example 3.34 — AI in criminal justice

- Example 3.34 — hands-on module
- Example 3.34 returns to recidivism and hotspot conflicts from earlier in the chapter
- A model may improve average institutional metrics while concentrating harm on already
- Justice criteria that inspect subgroup harms, not citywide averages alone
- View files: ``modules/chapter3/example34/``

Stakeholder involvement

- Section 3.1 and Example 3.1 named practitioners, leaders, policymakers, and data subjects
- Decision-making adds a process requirement
- Document who was consulted, which lens prevailed

Takeaways

- No single lens dissolves every conflict, teams should apply several deliberately
- Stakeholder review surfaces residual risks a single product owner may miss
- Accountability requires a record of the choice, not only the shipped feature

Next

- Complete the quiz for this part
- Continue to the final part on emerging trends, AI surveillance, global regulation friction

Learning objectives

- Describe AI-scale privacy risks, name two global regulation friction patterns
- Identify near-term directions including privacy-enhancing technologies and stronger

AI and privacy concerns

- Large-scale sensing and prediction create surveillance capacity that traditional
- Example 3.35 focuses on facial analysis at city scale

Example 3.35 — Surveillance and facial analysis

- Example 3.35 — hands-on module
- Example 3.35 connects technical accuracy gaps to civil-liberties risk
- Explore the chapter example module
- View files: ``modules/chapter3/example35/``

Ethics debates in deployed AI

- How much human control must remain in high-stakes loops
- Example 3.37 illustrates autonomous vehicle liability

Global privacy regulation challenges

- Cross-border products confront overlapping rules rather than a single global code
- Request-handling workflows
- Localization mandates and transfer restrictions

Example 3.38 — Conflicting privacy regimes

- Example 3.38 — hands-on module
- The operational cost of running multiple statutory programs in one product
- Explore the chapter example module
- View files: ``modules/chapter3/example38/``

Future directions

- Privacy by Design defaults
- Example 3.41 sketches self-sovereign identity proposals that let people hold and present

Example 3.41 — Self-sovereign identity

- Example 3.41 — hands-on module
- Example 3.41 aligns with stronger individual control over personal data
- Explore the chapter example module
- View files: ``modules/chapter3/example41/``

Looking ahead

- Emerging tensions
- Chapter 7 deepens bias measurement and mitigation
- Chapter 4 turns to annotation

Takeaways

- AI scale outpaces notice-only governance
- Global products need jurisdiction maps and transfer plans
- PETs and design defaults are mainstream, the adoption gap is organizational

Next

- Complete the quiz for this part
- Pause for the quiz